Text-driven image style transfer: Achieving precise visual manipulation through Stable Diffusion Model, Prompt-to-Prompt, and Null-text Inversion

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1 Introduction

In recent years, with the rapid development of generative artificial intelligence technology, text-guided image generation models (such as Stable Diffusion[1]) have demonstrated great potential in fields like content creation, digital art, and advertising design. However, how to conduct precise and controllable semantic-level editing on existing images (including real photos) remains a key challenge in current research. Traditional methods (such as GAN-based image editing[2]) often struggle to preserve the structural details of the original image while modifying the style or content. For instance, when changing the color of a red apple picture to green, it might accidentally alter its shape, position, or even the background layout. Therefore, there is an urgent need for a new paradigm that can not only maintain pixel-level content consistency but also achieve flexible semantic style transfer. This project focuses on text-driven image style transfer, aiming to integrate cutting-edge technologies such as Stable Diffusion, Prompt-to-Prompt (P2P)[3], and Null-Text Inversion (NTI)[4] to construct a high-fidelity, editable, and fine-tuning-free image manipulation framework. This research not only promotes the development of generative models in the direction of controllable editing, but also provides effective tools for practical applications such as digital content creation and personalized image processing.

2 Research objectives

Our goal is to integrate P2P and NTI methods under the Stable Diffusion architecture to achieve high-fidelity reconstruction and precise style transfer of real images. Specific research objectives include: achieving high-quality inversion of real images to enable them to be embedded in the latent space of diffusion models; By using prompt-to-prompt technology, the local or global style of an image can be controlled by modifying only the text prompt. The reconstruction quality is optimized by means of Null-Text Inversion, so that the edited image not only conforms to the new style semantics but also strictly retains the original structure. The main work content is as follows:

Step 1: Dataset Acquisition and Exploration Collect and organize multiple types of image data, including natural scene images from the COCO val2017 dataset and self-collected real photos. Conduct a preliminary analysis of the structural complexity, semantic clarity and style diversity of various types of images.

Step 2: Data preprocessing and experimental environment construction: Standardize the images (such as size unification and format conversion), and build a basic experimental platform based on Stable Diffusion.

Step 3: Model implementation and integration of key technologies to achieve Null-Text Inversion (NTI), completing the high-fidelity mapping from real images to potential noise; Integrate the Prompt-to-Prompt (P2P) mechanism to achieve spatial layout maintenance by injecting the cross-attention maps of the source image.

Step 4: Model Evaluation and Effect Comparison: The system performance is evaluated by combining quantitative and qualitative methods:

- DINO Feature similarity: Measures the consistency of object posture and layout before and after editing (the higher, the better);
- LPIPS perceived distance: Evaluate the naturalness of image details and the degree of artifacts (the lower, the better);
- CLIP Semantic alignment score: Determine whether the generated result conforms to the semantics of the new prompt word (such as "ink wash painting", "cartoon style", etc., the higher the better)[5].

Meanwhile, the editing effects were compared on different data subsets to verify the generalization ability of the method.

3 Methods

3.1 Stable Diffusion Model

The core of the diffusion model is divided into two stages: the forward process and the revers process. The forward process involves gradually adding random noise to the original image until it becomes pure noise. This process is fixed, and the goal is to train a noise prediction network. The reverse process starts from pure noise and utilizes the trained network to predict and remove noise at each step, gradually restoring a clear image.

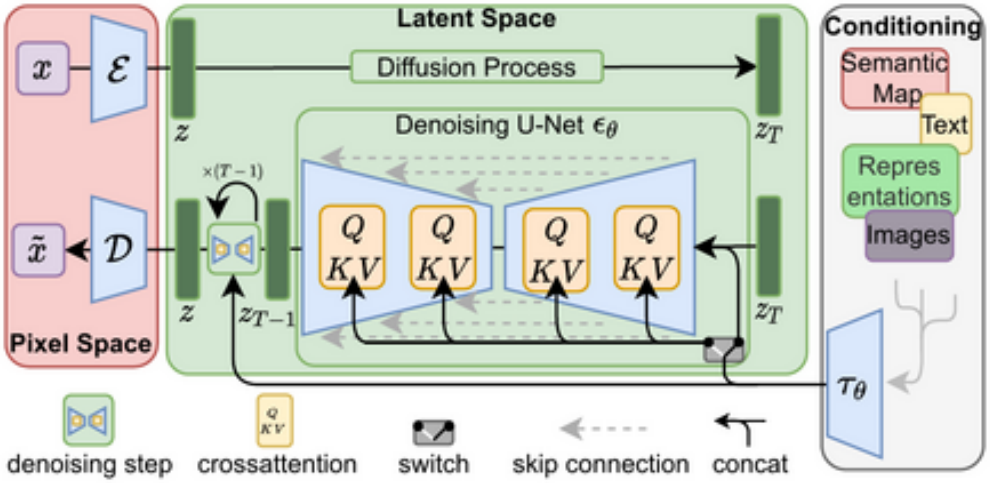


Figure 1: Stable Diffusion Model Framework

In our project, we select the Stable Diffusion Model[1], which is a latent-variable-based generative diffusion model. Its core advantage is lies in shifting the diffusion process from high-dimensional pixel space to low-dimensional latent space, thereby significantly reducing computing overhead. The basic framework of the SD Model is shown in the figure. The Variational Autoencoder (VAE) is responsible for the bidirectional mapping between image space and latent space; the encoder compresses the input image into a latent vector, while the decoder is responsible for restoring the processed latent vector into a pixel image. The conditional U-Net is the core network in the denoising stage; it learns to predict the noise within the latent variables and is guided by combing text features from CLIP[5]. The text encoder transforms the input text prompts into semantic embedding vectors, which act on the U-Net through the cross-attention mechanism to achieve precise semantic control over the generated content.

3.2 Prompt-to-Prompt

In editing tasks based on Stable Diffusion, traditional methods (such as Inpainting or SDEdit[6]) often face a dilemma: either they require cumbersome manual masking, which severely compromises the convenience and intuitiveness of text-driven image editing, or they destroy the original composition structure when modifying prompts, failing to ensure consistency between the image background and the original content (See Figure2). This project selects Prompt-to-Prompt as the fundamental architecture, whose core advantage lies in the fact that it does not require model retraining, fine-tuning, or additional datasets and optimization processes. It can achieve pixel-level structural preservation by directly intervening in the attention layers during the generation process.



Figure 2: Content modification through attention injection

The core of Prompt-to-Prompt (P2P) technology[3] lies in manipulating the cross-attention layers within the U-Net architecture. This layer is responsible for merging the visual features of the image with the textual embedding features (See Figure3 top), generating a spatial attention map for each textual token. For the spatial features $\phi(z_t)$ and text embeddings c of the input image, the attention map M is calculated as follows:

$$M = Softmax(\frac{QK^T}{\sqrt{d}}), Output = MV$$

Where $Q = l_Q(\phi(z_t))$ represents the image spatial features, $K = l_K(c)$ represents the feature mapping of the text prompt words. The cross-attention output MV is a weighted average of the vectors of token values, which is used to update the spatial feature representation of the image.

The core algorithm of P2P simultaneously executes noise removal processes on both the original prompt P and the modified prompt P^* at each diffusion time step t. Through an attention map injection mechanism, the structural information of the original image is transferred to the edited image generation process:

1. Perform standard diffusion noise removal using the prompt P and random seed s, recording the attention map M_t generated at each step;
2. Perform diffusion noise removal using the modified prompt P^*, generating the attention map M_t^*;
3. Merge both attention maps using the edit function $Edit(M_t, M_t^*, t)$ to produce the controlled attention map M_t^c;
4. Replacing M_t^c with the original attention map in the P^* generation process enables semantic editing while preserving the structure of original image.

It should be noted that the injection of the attention map only replaces the attention weighting matrix M, preserving the value matrix V generated by the modified indication P^*. This ensures that the edited image retains its original geometric structure, while its pixel content responds to the semantics of the new indication. Furthermore, both diffusion processes share identical random seeds and initial noise, ensuring consistency in the generated results.

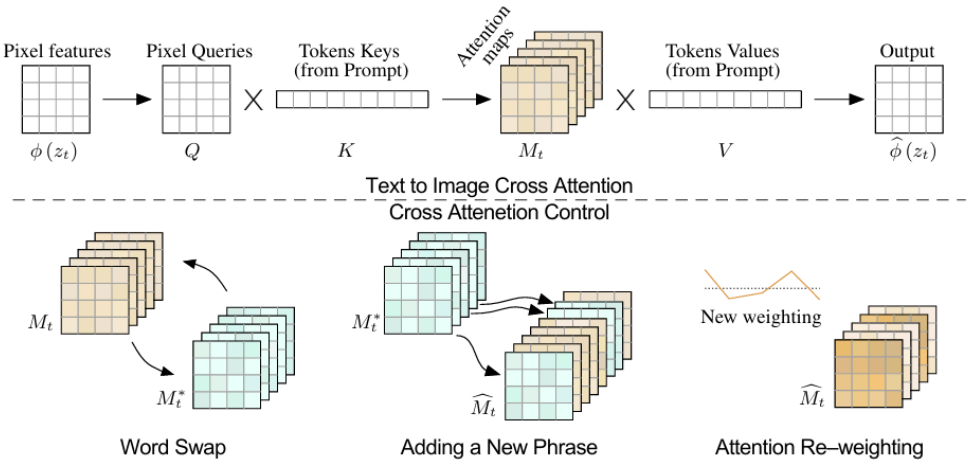


Figure 3: Prompt-to-Prompt method overview

We now turn to address specific editing operations, filling the missing definition of the $Edit(M_t, M_t^*, t)$ function. An overview is presented in Figure 3(Bottom).

- **Word Swap**: In this case, the user swaps tokens of the original prompt with others, e.g., $P =$ 'a big red bicycle' to $P^* =$ 'a big red car'. The main challenge is that the proposed attention injection may over constrain the geometry of new objects, resulting in the generated 'cars' still looking like 'bicycles'. We address this by suggesting a temporal step parameter τ as a softer attention constrain:

$$Edit(M_t, M_t^*, t) = \begin{cases} M_t^*, & \text{if } t < \tau \\ M_t, & \text{if } t \geq \tau \end{cases}$$

Specifically, during the first diffusion steps ($t \geq \tau$, the formation of the general image design), the attention map of the original image is injected to preserve structural integrity. In later steps ($t < \tau$, the detail generation phase), the attention map of the new indication is used to grant semantic freedom. A larger τ produces results closer to the original image, while a smaller τ allows for a more receptive interpretation of the new indication. In addition, different injection step counts can be assigned to different tokens within the indication, allowing for more precise control.

- **Adding a New Phrase**: In another setting, the user adds new tokens to the prompt, e.g., $P =$ 'a castle next to a river' to $P^* =$ 'children drawing of a castle next to a river'. To preserve the common details, we apply the attention injection only over the common tokens from both prompts. Formally, we use an alignment function A that receives a token index from target prompt P^* and outputs the corresponding token index in P or None if there isn't a match. Then, the editing function is given by:

$$(Edit(M_t, M_t^*, t))_{i,j} = \begin{cases} M_{t\,i,j}^* & \text{if } A(j) = \text{None} \\ M_{t\,i,A(j)} & \text{otherwise} \end{cases}$$

  Specifically: for tokens shared between P^* and P, the attention map of the original image is used (preserving the structure); for newly introduced tokens, the attention map generated by P^* itself is used (allowing it to respond to new content). This strategy preserves existing structural elements while allowing the semantic influence of the new vocabulary to be integrated naturally, thus achieving overall stylization or the addition of local attributes.

- **Attention Re-weighting**: Lastly, the user may wish to strengthen or weakens the extent to which each token is affecting the resulting image. For example, consider the prompt $P =$ 'a fluffy red ball', and assume we want to make the ball more or less fluffy. To achieve such manipulation, we scale the attention map of the assigned token j^* with parameter $c \in [2, 2]$, resulting in a stronger/weaker effect. The rest of the attention maps remain unchanged. That is:

$$(Edit(M_t, M_t^*, t))_{i,j} = \begin{cases} c \cdot M_{t\,i,j}^* & \text{if } j = j^* \\ M_{t\,i,j} & \text{otherwise.} \end{cases}$$

3.3 Null-text Inversion

Although P2P performs excellently on generated images, its application to real images faces limitations. Because the DDIM[7] reconstruction used in P2P involves a 'distortion-editability tradeoff', the model often fails to accurately reconstruct the original image when applied directly to real images. Therefore, this project further incorporates Null-Text Inversion (NTI) technology for enhancement.

Null-text Inversion (NTI) is a workflow designed for real image editing[4][8]. Its operation can be summarized as embedding real photographs into the latent space of diffusion models, followed by flexible modifications through text prompts. Specifically, we only need to provide an input image along with a textual description. NTI then applies pivotal inversion to obtain a stable initial trajectory, and builds upon it with null-text optimization to achieve high-fidelity reconstruction. After this process, editing becomes straightforward: simply change the prompt to perform modificationsfor example, turning 'cat' into 'tiger' for object replacement, 'photo' into 'sketch' for style transformation, or 'flowers' into 'cupcakes' for semantic substitution. Importantly, the optimized embedding obtained from a single inversion can be reused, supporting multiple different Prompt-to-Prompt edits without retraining or fine-tuning the model.

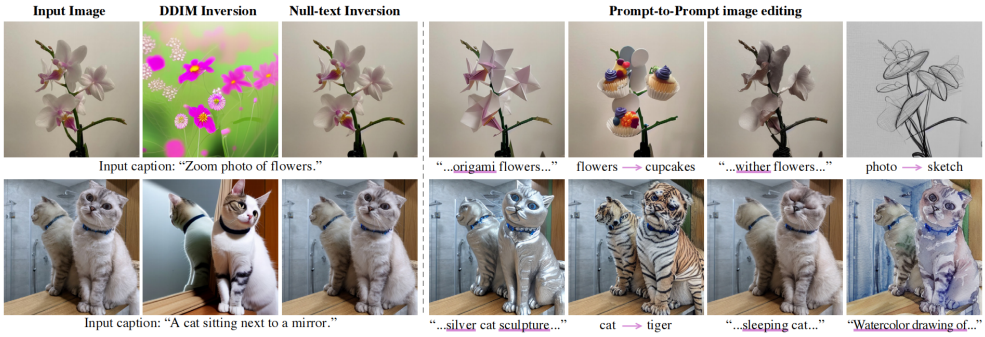


Figure 4: Null-text Inversion for real image editing

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